BRIEFING

Fine-tuning, LoRA, and QLoRA

How a pretrained language model learns a specific job, and two ways to train it without touching every weight.

Fine-tuning

Keep training on your own examples.

LoRA

A small adapter on a 16-bit model.

QLoRA

The same adapter, with a 4-bit model.

What fine-tuning is

A pretrained model already knows general language. Fine-tuning keeps training it on a smaller set of examples that look like the job you want. The saved weights change, so the new behavior stays with the model.

Prompting

You change the words sent with each request. The saved model stays as it was.

Retrieval

You attach source documents at answer time. The saved model stays as it was.

Fine-tuning

You continue training. The saved weights carry the new behavior.

EXAMPLE

Ask: Describe a matte black desk lamp.

Before: This stunning lamp will transform your workspace and elevate every evening.

After training on calm product lines: A matte black desk lamp with a low arm and a warm, even pool of light. It stays out of the way and lights the page.

Which parameters move

Full fine-tuning updates every weight. Parameter-efficient fine-tuning freezes those weights and trains a small add-on. LoRA and QLoRA are the second kind.

EXAMPLE — FULL FINE-TUNING

A 7-billion-parameter model has about 7 billion trainable numbers.
You update all of them on your support chats.
You save a new full model, about 14 GB in 16-bit.

EXAMPLE — PARAMETER-EFFICIENT

The same 7 billion numbers stay frozen.
You train an adapter of a few million numbers instead.
You save the adapter, often tens of megabytes, and keep the original model.

Prefix tuning and bottleneck adapters are other parameter-efficient methods. This briefing stays with LoRA and QLoRA.

What the examples teach

The training signal is a separate choice from how many weights move. The same LoRA run can use any of these data styles. This project uses the first one.

SUPERVISED FINE-TUNING

User: Describe a linen notebook cover.

Assistant: A linen notebook cover in a natural weave. It takes on a soft crease with use.

A prompt paired with the answer you want. Instruction tuning is this, on instruction-style pairs.

CONTINUED PRETRAINING

The cover cloth is linen, 240 gsm, sewn onto boards. The thread is cotton. The endpaper is left blank.

More raw text from a field. There is no question and no labeled answer.

PREFERENCE TUNING

Ask: Describe a wool throw.

Chosen: A wool throw in a plain weave. It folds flat at the end of a sofa.

Rejected: This amazing blanket will change your life.

The model is trained to prefer the answer a person would choose. DPO uses pairs like this. RLHF uses a reward score.

A small update beside a frozen layer

Low-Rank Adaptation leaves the original weight matrix W frozen. It trains two thin matrices, A and B, whose product stands in for a change to that layer. The rank r is the thin dimension. Here r = 16.

PIECE	ROLE	SHAPE	VALUES
W	Original layer, frozen, 16-bit	1024 × 1024	1,048,576
A	Trained	1024 × 16	16,384
B	Trained	16 × 1024	16,384

The layer still computes a 1024 × 1024 transform. Training touches 32,768 numbers, about 3% of W. The forward pass adds the adapter: output = Wx + BAx.

```
python train.py --method lora --train-file data/sample_train.jsonl
```

The same adapter, with W stored in 4-bit

QLoRA keeps the LoRA example above and quantizes the frozen matrix. A and B still train in 16-bit. Only the stored base weights shrink.

STORAGE OF W	BITS PER VALUE	BYTES FOR THIS LAYER
LoRA base	16	2,097,152 bytes (2.0 MB)
QLoRA base, NF4	4	524,288 bytes (0.5 MB)
Adapter A and B, either method	16	65,536 bytes (64 KB)

On a full 7B model the same ratio applies to the weights: about 14 GB in 16-bit, about 4 GB in 4-bit, before activations and the optimizer. NormalFloat4 is the 4-bit type. Double quantization compresses the scale constants a second time.

```
python train.py --method qlora --train-file data/sample_train.jsonl
```

06 — CHOOSE

LoRA or QLoRA

Same supervised examples. Same adapter. The choice is whether the frozen base model is stored in 16-bit or 4-bit.

	FULL FINE-TUNING	LORA	QLORA
Trained	All weights	Adapter only	Adapter only
Base precision	16-bit	16-bit	4-bit
Saved result	Full model	Adapter	Adapter
Use when	Compute is plentiful	The 16-bit model fits	You need a larger model on one GPU

EXAMPLE — LORA

Qwen2.5-0.5B-Instruct fits easily in 16-bit. Train the adapter there when you want the frozen weights at full precision.

EXAMPLE — QLORA

Qwen2.5-7B-Instruct is the case for 4-bit. The adapter is unchanged. The base is what gets smaller.